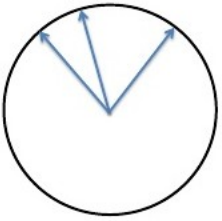
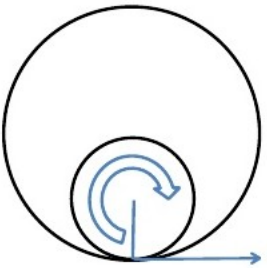
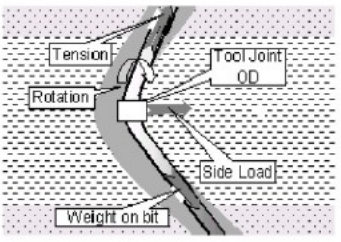
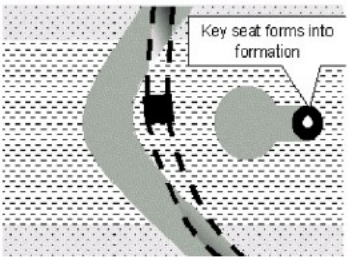
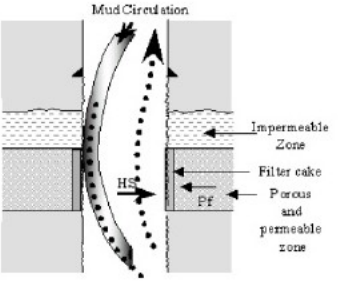

Vertical well
Horizontal well
Directional well

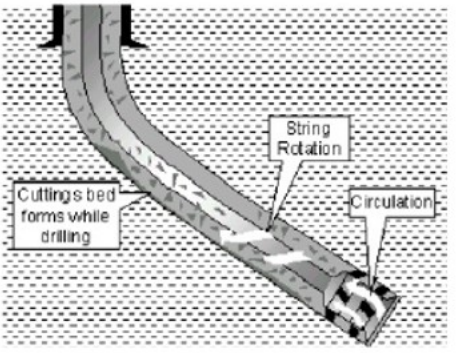


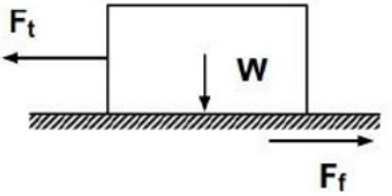


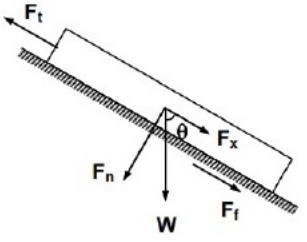

Tension
Tool Joint OD
Rotation
Side Load
Weight on bit

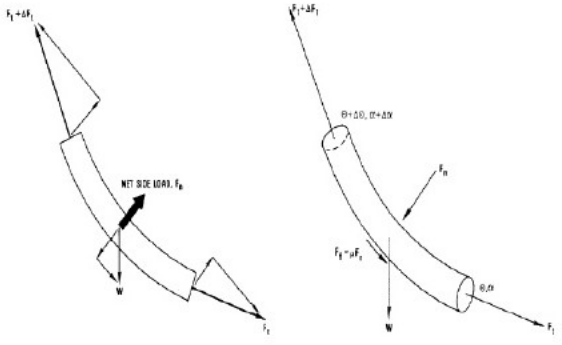

Key seat forms into formation

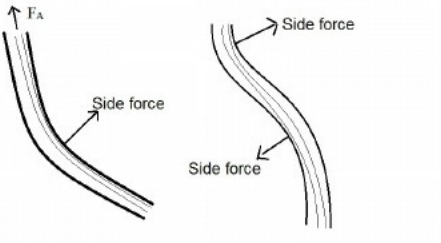

Mud Circulation
Impermeable Zone
Filter cake
Porous and permeable zone
HS
Pf

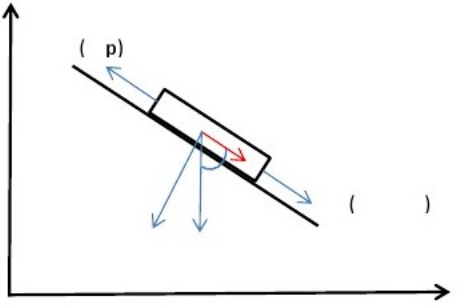

String Rotation
Circulation
Cuttings bed forms while drilling

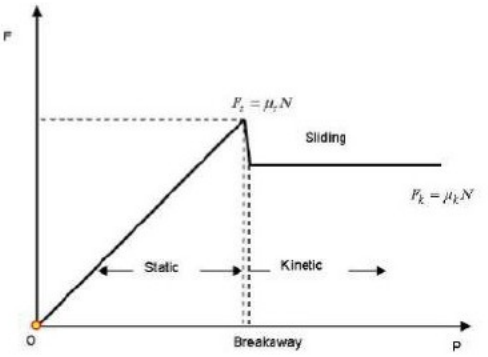

F_t
W
F_f

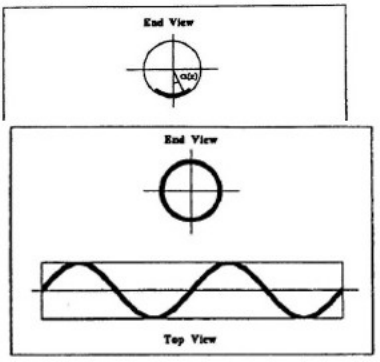

F_t
θ
F_x
F_n
F_f
W


$F_1+\Delta F_1$
NET SIDE LOAD, F_n
W
F_1
$F_1+\Delta F_1$
$\theta+\Delta\theta,\ x+\Delta x$
F_n
$F_f=\mu F_n$
W
θ, x
F_1

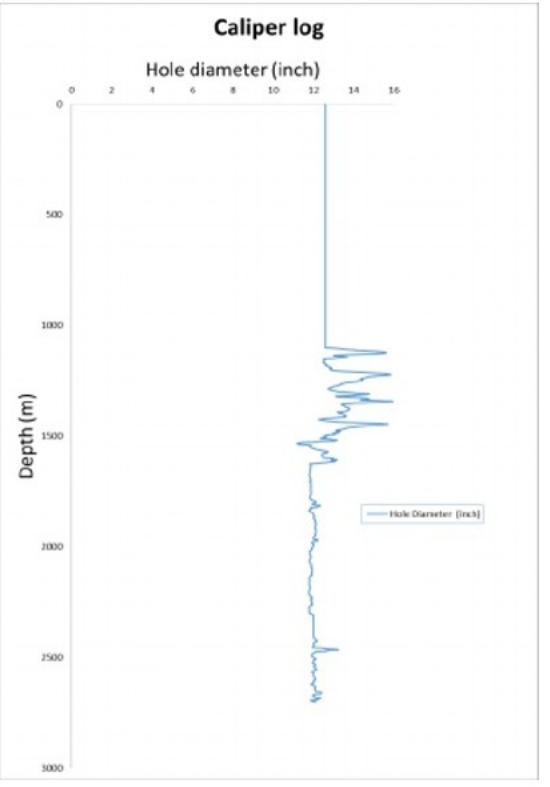

F_A
Side force
Side force
Side force

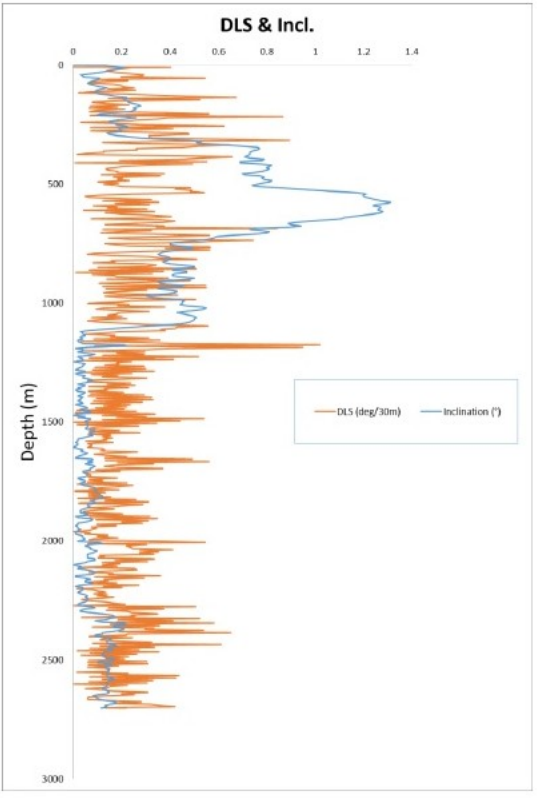

(p)
( )

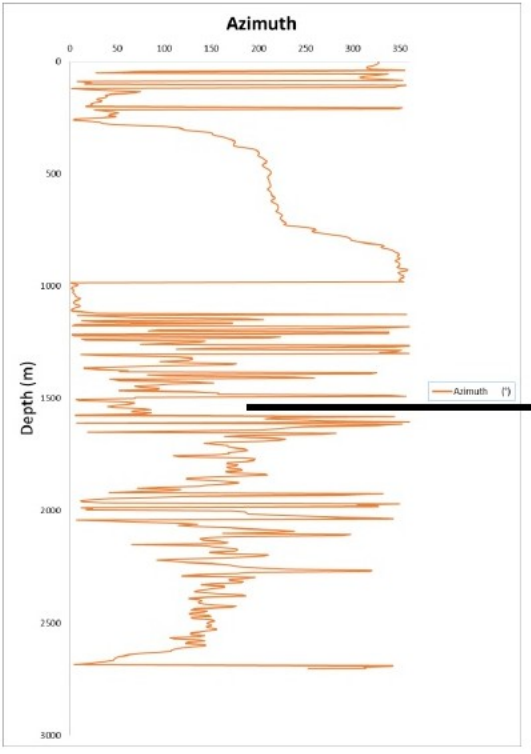

F
$F_s = \mu_s N$
Sliding
$F_k = \mu_k N$
Static
Kinetic
O
Breakaway
P

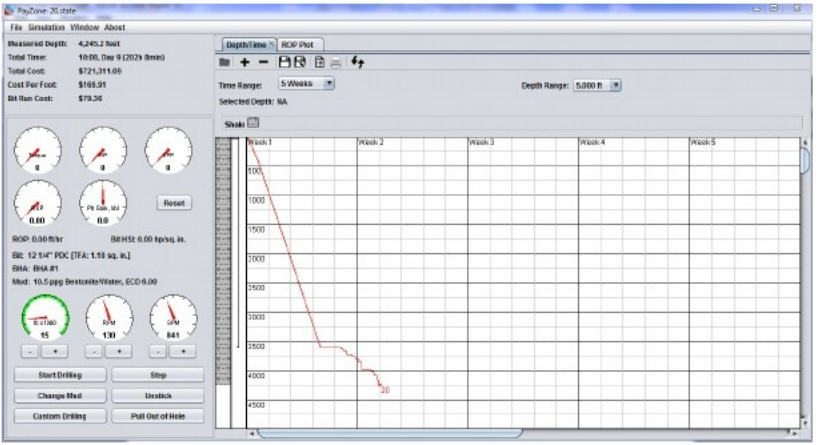

End View
End View
Top View

12.1/4" Hole assembly	Body OD	Connections	Supplier
12-1/4" Bit PDC SMITH Bit XS616 Nozzles: 6x14	12.250	6.625 REG Pin	SMITH
Power-V – PD900 x5 w/ 12-1/8" stab	9.125	6.625 REG B X 6.625 REG B	SLB
8-1/4" Collar	8.250	6.625 REG P X 6.625 REG B	Rig Contractor
12-1/8" Stabilizer	12.125	6.625 REG P X 6.625 REG B	SinoGulf
Float sub (with non- ported float)	8.5	6.625 REG P X 6.625 REG B	SinoGulf
WINC	8.000	6.625 REG P X 6.625 REG B	MidWest
10 X 8-1/4" Collar	8.250	6.625 REG P X 6.625 REG B	Rig Contractor
Dogleg reamer (DLR)	8.250	6.625 REG P X 6.625 REG B	NOV
3 X 8-1/4" Collar	8.250	6.625 REG P X 6.625 REG B	Rig Contractor
8" Hydro mechanical Drilling Jar	8.000	6.625 REG P X 6.625 REG B	SinoGulf
3 X 8-1/4" Collar	8.250	6.625 REG P X 6.625 REG B	Rig Contractor
Crossover	8.250	6.625 REG P X 4.5 NC50 (4 1/2 IF) B	Rental
9 X 6-3/4" Collar	6.750	NC50 P X B	Rig Contractor
15 X 5" HWDP	5.000	NC50 P X B	Rig Contractor


Caliper log
Hole diameter (inch)
0
2
4
6
8
10
12
14
16
Depth (m)
0
500
1000
1500
2000
2500
3000
Hole Diameter (inch)

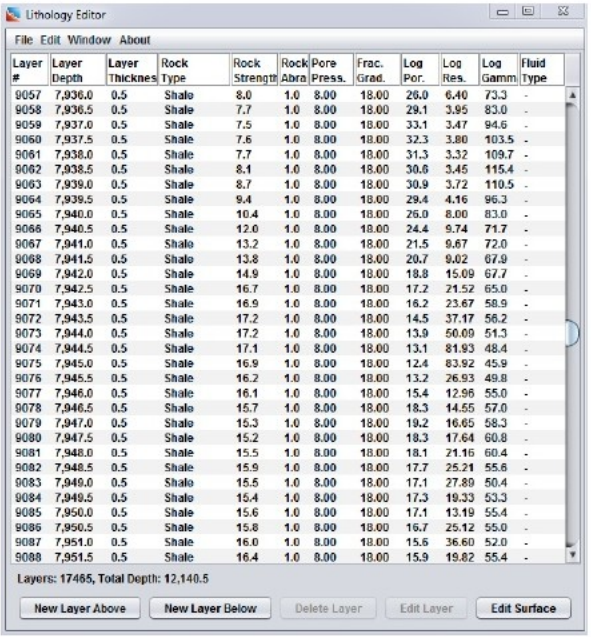

DLS & Incl.
0
0.2
0.4
0.6
0.8
1
1.2
1.4
0
500
1000
1500
2000
2500
3000
Depth (m)
DLS (deg/30m)
Inclination (°)


Azimuth
0
50
100
150
200
250
300
350
0
500
1000
1500
2000
2500
3000
Depth (m)
Azimuth (°)

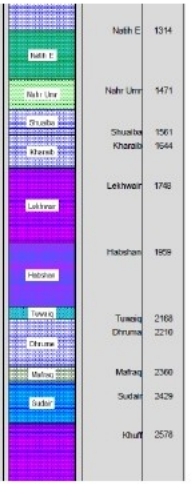

PayZone 26.state
File Simulation Window About
Measured Depth: 4,245.2 feet
Total Time: 10:08, Day 9 (202h 8min)
Total Cost: $721,311.06
Cost Per Foot: $169.91
Bit Run Cost: $79.36
Reset
ROP: 0.00 ft/hr
Bit HSI: 0.00 hp/sq. in.
Bit: 12 1/4" PDC [TFA: 1.10 sq. in.]
BHA: BHA #1
Mud: 10.5 ppg Bentonite/Water, ECD 0.00
WOB
95
RPM
130
GPM
841
Start Drilling
Stop
Change Mud
Unstick
Custom Drilling
Pull Out of Hole
Depth/Time
ROP Plot
Time Range:
5 Weeks
Depth Range:
5,000 ft
Selected Depth: NA
Shale
Week 1
Week 2
Week 3
Week 4
Week 5
500
1000
1500
2000
2500
3000
3500
4000
4500
20

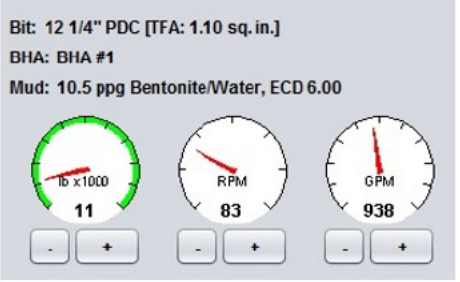
LAS column title	PayZone parameter
DEPT	depth
GR	gamma ray emission rate
ILD	resistivity (Induction Log Deep)
NPHI	porosity (inactive)
PORO	porosity
SP	spontaneous potential
RHOB	density
DT	compressive wave sonic velocity
ST	shear wave sonic velocity
UCS	unconfined compressive strength
ABR	rock abrasivity
FRICANG	angle of internal friction
PPRE	pore pressure gradient
FRAC	fracture pressure gradient
V_CO	volume fraction of coal
V_DO	volume fraction of dolomite
V_LS	volume fraction of limestone
V_HS	volume fraction of hard sandstone
V_SS	volume fraction of soft sandstone
V_SH	volume fraction of shale
V_TF	volume fraction of tuff

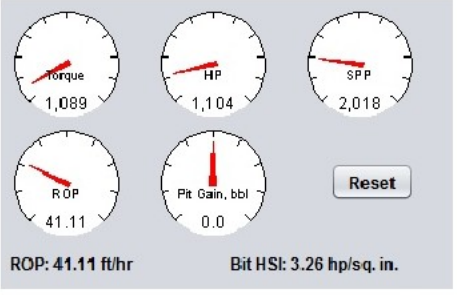
#LOGNAME	.UNIT	LOG_ID
#		
MD	.m	
DT	.us/ft	
GR	.gAPI	
PORO	.v/v	
ILD	.ohm.m	
RHOB	.g/cm3	
UCS	.psi	

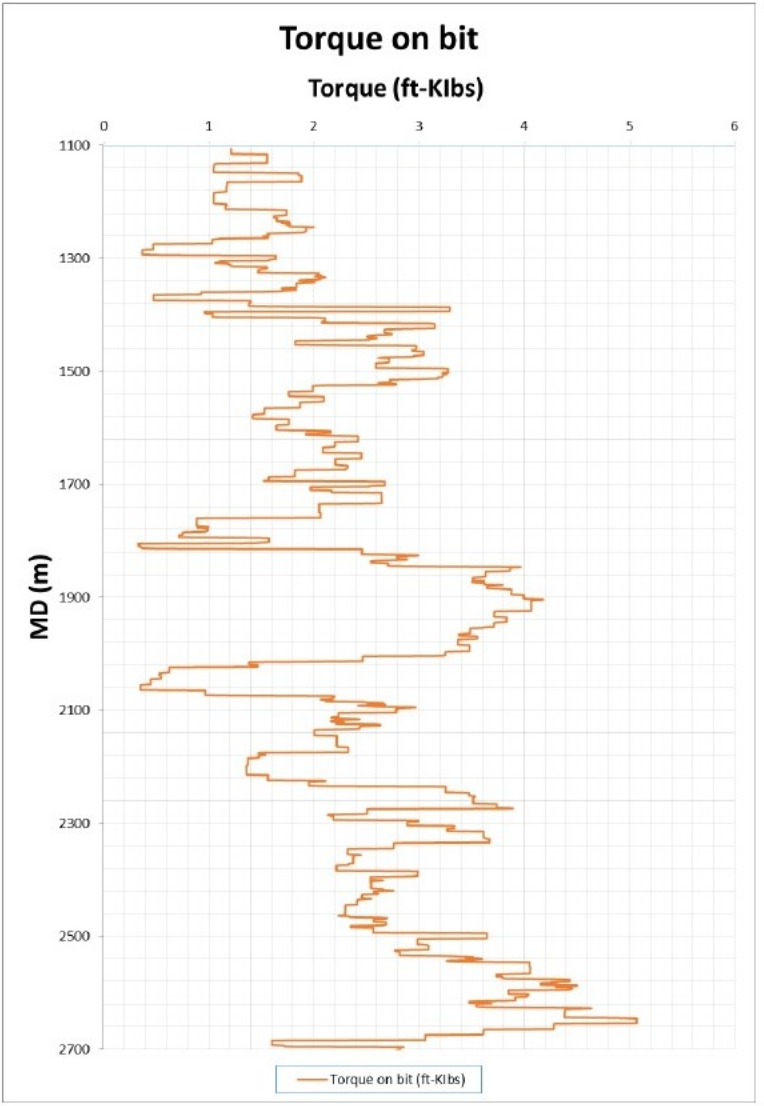


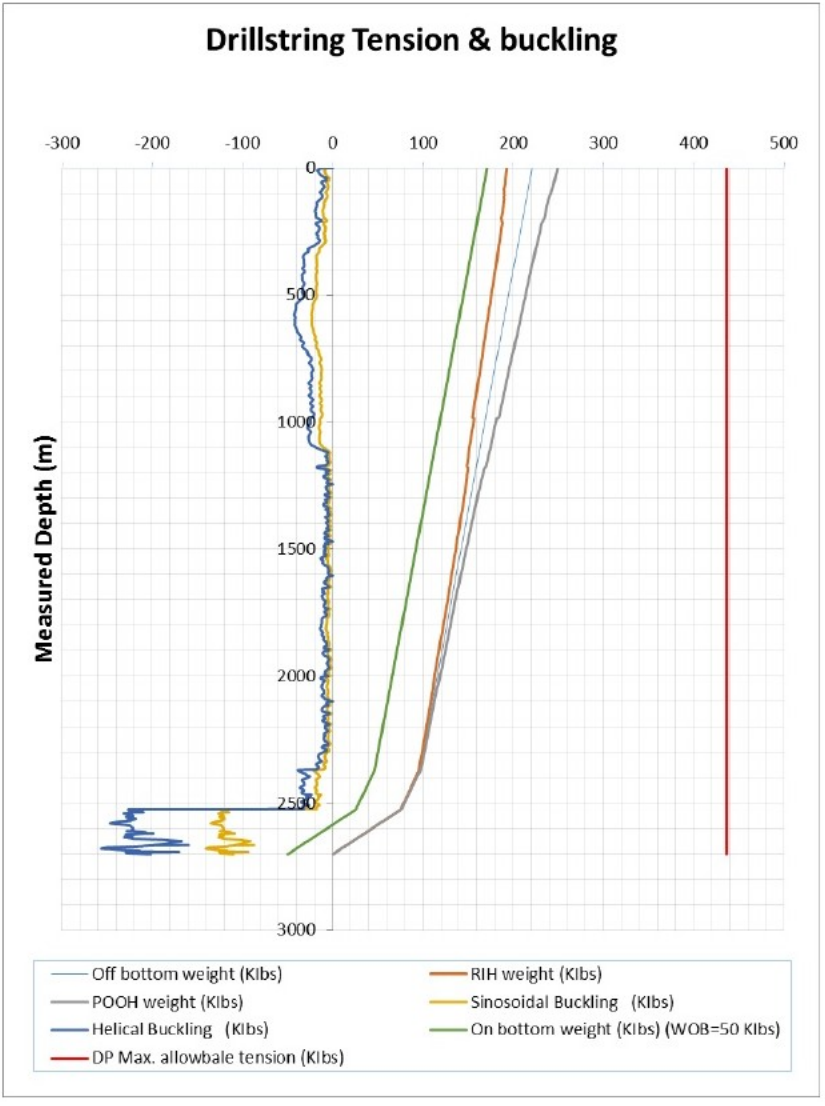
Layer #	Layer Depth	Layer Thickness	Rock Type	Rock Strength	Rock Abra	Pore Press.	Frac. Grad.	Log Por.	Log Res.	Log Gamm	Fluid Type
9057	7,936.0	0.5	Shale	8.0	1.0	8.00	18.00	26.0	6.40	73.3	-
9058	7,936.5	0.5	Shale	7.7	1.0	8.00	18.00	29.1	3.95	83.0	-
9059	7,937.0	0.5	Shale	7.5	1.0	8.00	18.00	33.1	3.47	94.6	-
9060	7,937.5	0.5	Shale	7.6	1.0	8.00	18.00	32.3	3.80	103.5	-
9061	7,938.0	0.5	Shale	7.7	1.0	8.00	18.00	31.3	3.32	109.7	-
9062	7,938.5	0.5	Shale	8.1	1.0	8.00	18.00	30.6	3.45	115.4	-
9063	7,939.0	0.5	Shale	8.7	1.0	8.00	18.00	30.9	3.72	110.5	-
9064	7,939.5	0.5	Shale	9.4	1.0	8.00	18.00	29.4	4.16	96.3	-
9065	7,940.0	0.5	Shale	10.4	1.0	8.00	18.00	26.0	8.00	83.0	-
9066	7,940.5	0.5	Shale	12.0	1.0	8.00	18.00	24.4	9.74	71.7	-
9067	7,941.0	0.5	Shale	13.2	1.0	8.00	18.00	21.5	9.67	72.0	-
9068	7,941.5	0.5	Shale	13.8	1.0	8.00	18.00	20.7	9.02	67.9	-
9069	7,942.0	0.5	Shale	14.9	1.0	8.00	18.00	18.8	15.09	67.7	-
9070	7,942.5	0.5	Shale	16.7	1.0	8.00	18.00	17.2	21.52	65.0	-
9071	7,943.0	0.5	Shale	16.9	1.0	8.00	18.00	16.2	23.67	58.9	-
9072	7,943.5	0.5	Shale	17.2	1.0	8.00	18.00	14.5	37.17	56.2	-
9073	7,944.0	0.5	Shale	17.2	1.0	8.00	18.00	13.9	50.09	51.3	-
9074	7,944.5	0.5	Shale	17.1	1.0	8.00	18.00	13.1	81.93	48.4	-
9075	7,945.0	0.5	Shale	16.9	1.0	8.00	18.00	12.4	83.92	45.9	-
9076	7,945.5	0.5	Shale	16.2	1.0	8.00	18.00	13.2	26.93	49.8	-
9077	7,946.0	0.5	Shale	16.1	1.0	8.00	18.00	15.4	12.96	55.0	-
9078	7,946.5	0.5	Shale	15.7	1.0	8.00	18.00	18.3	14.55	57.0	-
9079	7,947.0	0.5	Shale	15.3	1.0	8.00	18.00	19.2	16.65	58.3	-
9080	7,947.5	0.5	Shale	15.2	1.0	8.00	18.00	18.3	17.64	60.8	-
9081	7,948.0	0.5	Shale	15.5	1.0	8.00	18.00	18.1	21.16	60.4	-
9082	7,948.5	0.5	Shale	15.9	1.0	8.00	18.00	17.7	25.21	55.6	-
9083	7,949.0	0.5	Shale	15.5	1.0	8.00	18.00	17.1	27.89	50.4	-
9084	7,949.5	0.5	Shale	15.4	1.0	8.00	18.00	17.3	19.33	53.3	-
9085	7,950.0	0.5	Shale	15.6	1.0	8.00	18.00	17.1	13.19	55.4	-
9086	7,950.5	0.5	Shale	15.8	1.0	8.00	18.00	16.7	25.12	55.0	-
9087	7,951.0	0.5	Shale	16.0	1.0	8.00	18.00	15.6	36.60	52.0	-
9088	7,951.5	0.5	Shale	16.4	1.0	8.00	18.00	15.9	19.82	55.4	-

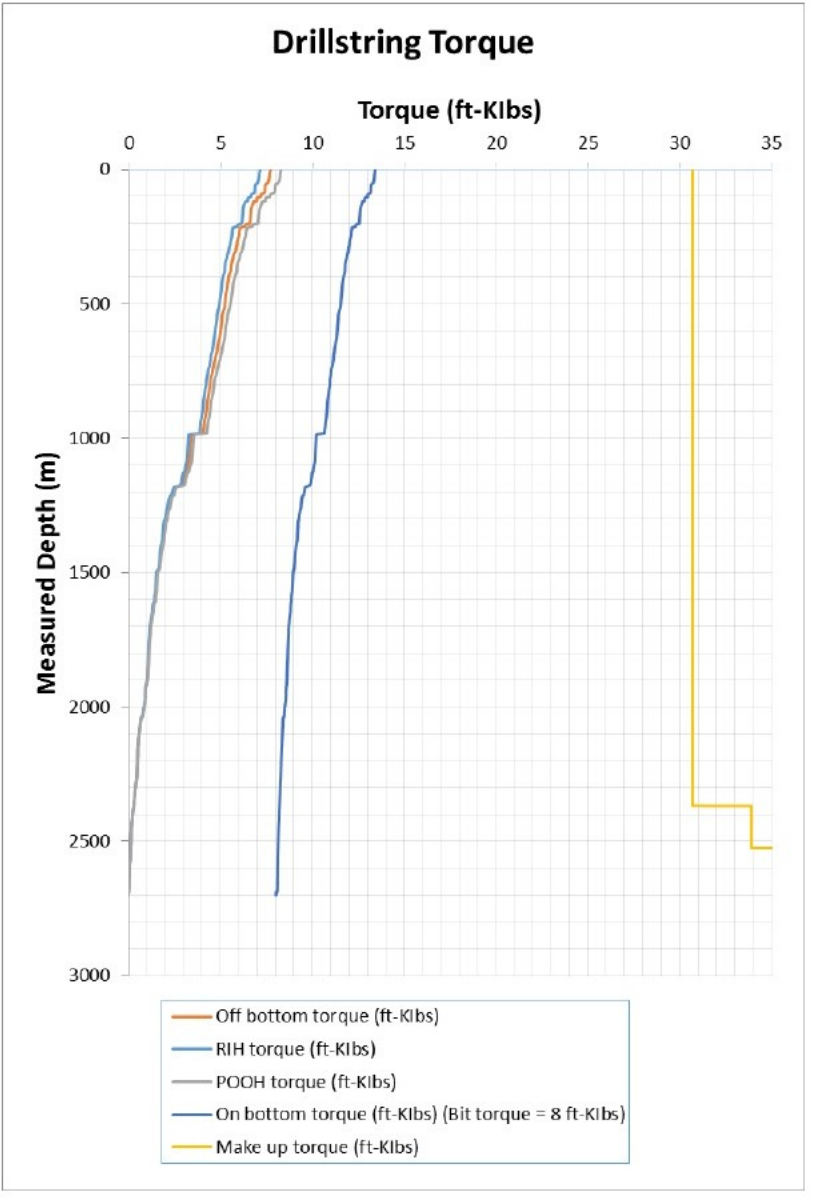

Nath E
1314
Nahr Umr
1471
Shuaiba
1561
Kharaib
1644
Lekhwair
1748
Habshan
1859
Tuwaiq
2168
Dhruma
2210
Mafraq
2360
Sudair
2429
Khuff
2578

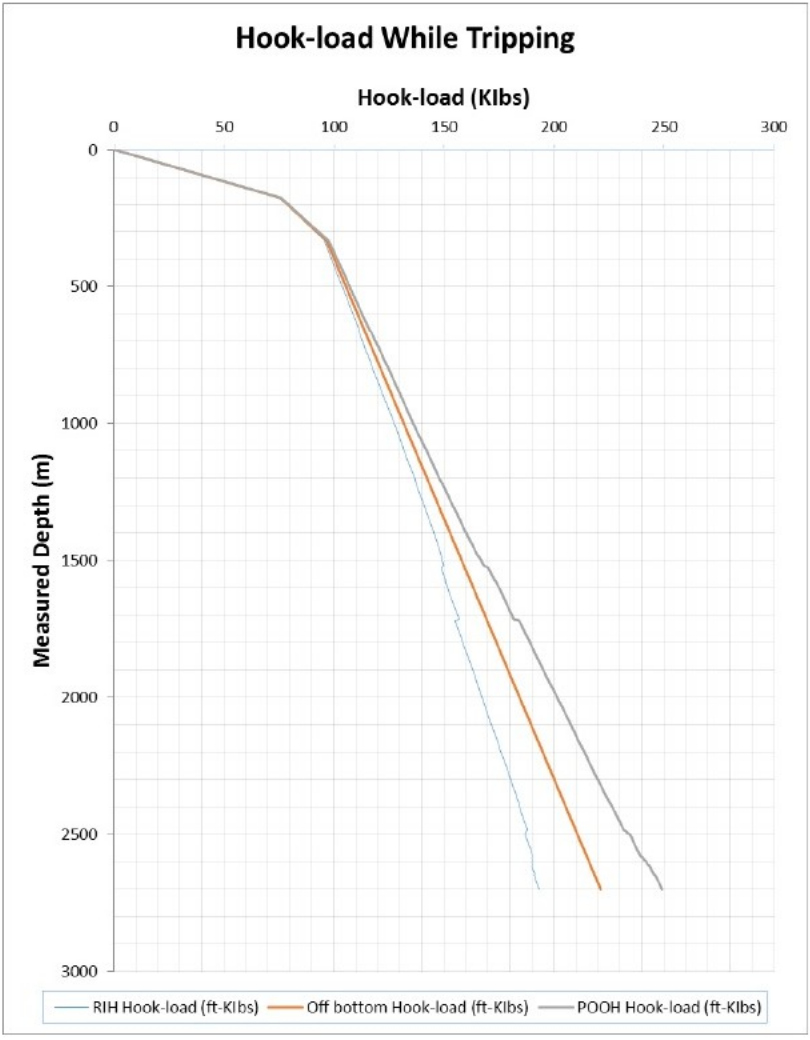

Bit: 12 1/4" PDC [TFA: 1.10 sq. in.]
BHA: BHA #1
Mud: 10.5 ppg Bentonite/Water, ECD 6.00
lb x1000
11
RPM
83
GPM
938
-
+
-
+
-
+

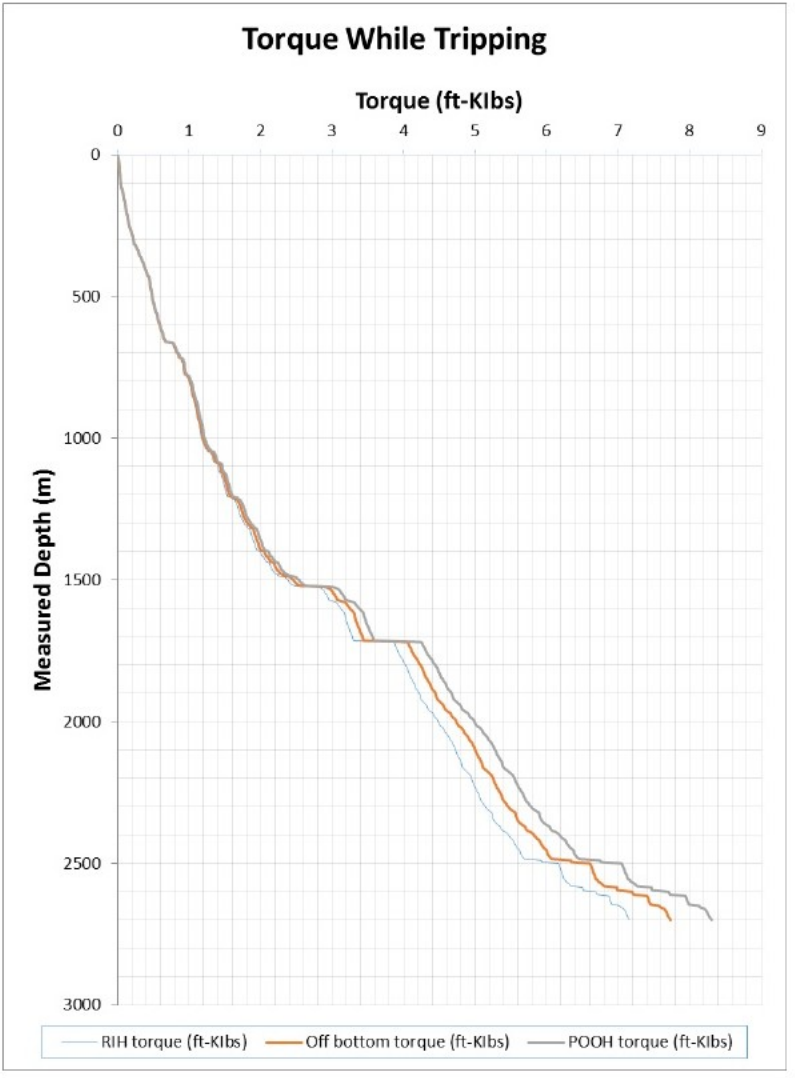

Torque
1,089
HP
1,104
SPP
2,018
ROP
41.11
Pit Gain, bbl
0.0
Reset
ROP: 41.11 ft/hr
Bit HSI: 3.26 hp/sq. in.

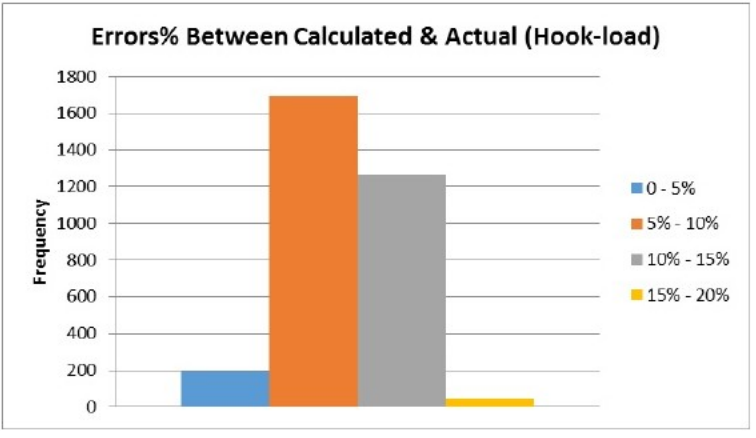

Torque on bit
Torque (ft-KIbs)
0
1
2
3
4
5
6
MD (m)
1100
1300
1500
1700
1900
2100
2300
2500
2700
Torque on bit (ft-KIbs)

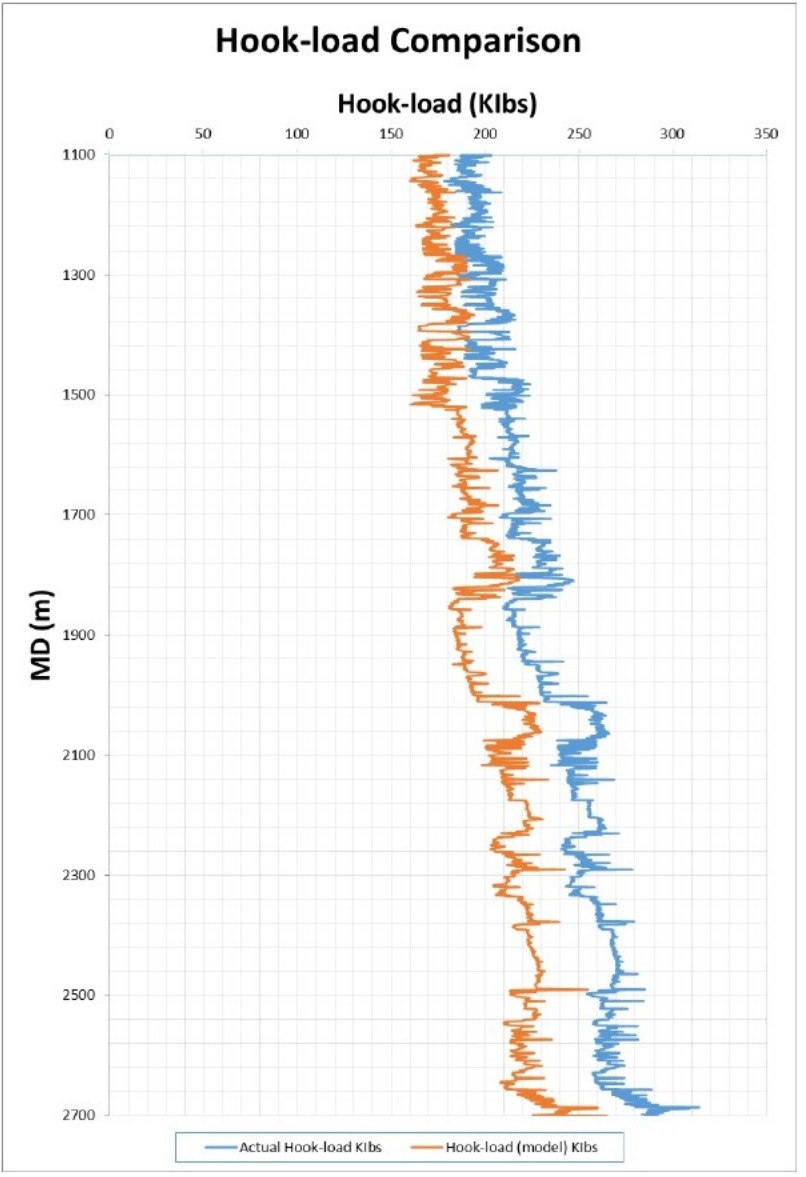

Drillstring Tension & buckling
-300
-200
-100
0
100
200
300
400
500
0
500
1000
1500
2000
2500
3000
Measured Depth (m)
Off bottom weight (Klbs)
RIH weight (Klbs)
POOH weight (Klbs)
Sinosoidal Buckling (Klbs)
Helical Buckling (Klbs)
On bottom weight (Klbs) (WOB=50 Klbs)
DP Max. allowbale tension (Klbs)

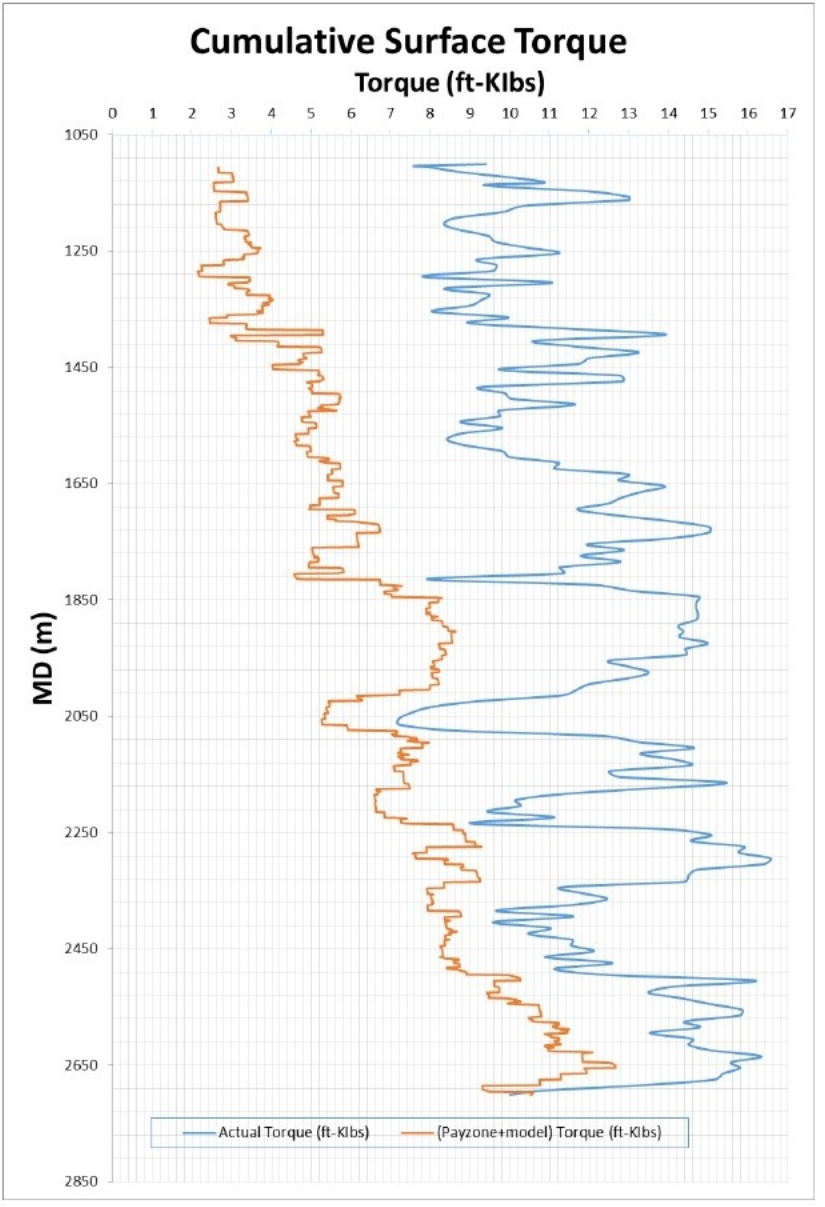

Drillstring Torque
Torque (ft-KIbs)
0
5
10
15
20
25
30
35
Measured Depth (m)
0
500
1000
1500
2000
2500
3000
Off bottom torque (ft-KIbs)
RIH torque (ft-KIbs)
POOH torque (ft-KIbs)
On bottom torque (ft-KIbs) (Bit torque = 8 ft-KIbs)
Make up torque (ft-KIbs)


Hook-load While Tripping
Hook-load (KIbs)
0
50
100
150
200
250
300
Measured Depth (m)
0
500
1000
1500
2000
2500
3000
RIH Hook-load (ft-KIbs)
Off bottom Hook-load (ft-KIbs)
POOH Hook-load (ft-KIbs)


Torque While Tripping
Torque (ft-KIbs)
0
1
2
3
4
5
6
7
8
9
Measured Depth (m)
0
500
1000
1500
2000
2500
3000
RIH torque (ft-KIbs)
Off bottom torque (ft-KIbs)
POOH torque (ft-KIbs)


Errors% Between Calculated & Actual (Hook-load)
Frequency
1800
1600
1400
1200
1000
800
600
400
200
0
0 - 5%
5% - 10%
10% - 15%
15% - 20%


Hook-load Comparison
Hook-load (Klbs)
0
50
100
150
200
250
300
350
MD (m)
1100
1300
1500
1700
1900
2100
2300
2500
2700
Actual Hook-load Klbs
Hook-load (model) Klbs


Cumulative Surface Torque
Torque (ft-Klbs)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
MD (m)
1050 1250 1450 1650 1850 2050 2250 2450 2650 2850
Actual Torque (ft-Klbs)
(Payzone+model) Torque (ft-Klbs)

